HOME-SHIELD

Home IoT Exposure Validator

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A defensive Python CLI tool for measuring and validating home/IoT network exposure through before-and-after testing.

Abstract

HOME-SHIELD is a Python-based command-line tool designed to measure exposure in home and small-office/home-office (SOHO) networks. The project focuses on a practical cybersecurity problem: home users often configure separate SSIDs, VLANs, guest networks, or firewall rules for IoT devices, but they do not always have a simple way to verify whether those changes actually reduced observable exposure. HOME-SHIELD addresses this gap by using a repeatable Measure -> Harden -> Re-measure workflow. The tool collects discovery and TCP reachability data, compares baseline and post-hardening runs, calculates an exposure score, and generates a readable HTML report.

The project was tested in a home-lab environment with multiple SSIDs, VLAN-style segmentation, a FortiWiFi 60F, an Xfinity ISP router, a macOS scanning host, and a mix of IoT and non-IoT devices. In the final scan, HOME-SHIELD compared baseline_iot_run1 against rerun_iot_run3. The exposure score improved from 55 to 58. The report showed that mDNS responders decreased from 9 to 6, open TCP ports decreased from 5 to 4, and two previously open services were closed. The tool also detected one newly open port, showing that it can report both improvements and regressions. Overall, HOME-SHIELD demonstrates that home network hardening should be measured rather than assumed, especially in SOHO environments where consumer routers may limit visibility and configuration control.

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1 - Introduction

1.A - Background

Home and SOHO networks have become more important to cybersecurity because they now support work, school, entertainment, smart-home automation, and personal data on the same local infrastructure. These networks are often administered by one person, usually the same person who depends on the network every day. Unlike enterprise networks, home networks rarely have dedicated monitoring, formal change control, or continuous security review.

Modern homes also contain far more than laptops and phones. A typical home network may include smart TVs, cameras, hubs, voice assistants, smart plugs, sensors, mobile devices, wearables, and other internet-connected systems. IoT Analytics projected that connected IoT devices would grow from 21.1 billion in 2025 to 39.0 billion in 2030 and 55.0 billion in 2035 [14]. This growth matters because each additional device can add discovery traffic, management interfaces, cloud connections, and local services to the home network.

IoT devices are usually designed around convenience. They advertise features such as remote access, smart-home control, automation, health monitoring, and local continuity between devices. These features are useful, but they also depend on the network being able to discover, reach, and exchange information with devices. When convenience is prioritized over security, users may end up with more visible services and more reachable endpoints than they realize. Earlier research has also shown that convenience, weak configuration, and exposed services can contribute to IoT security risks, including botnet-style abuse [11].

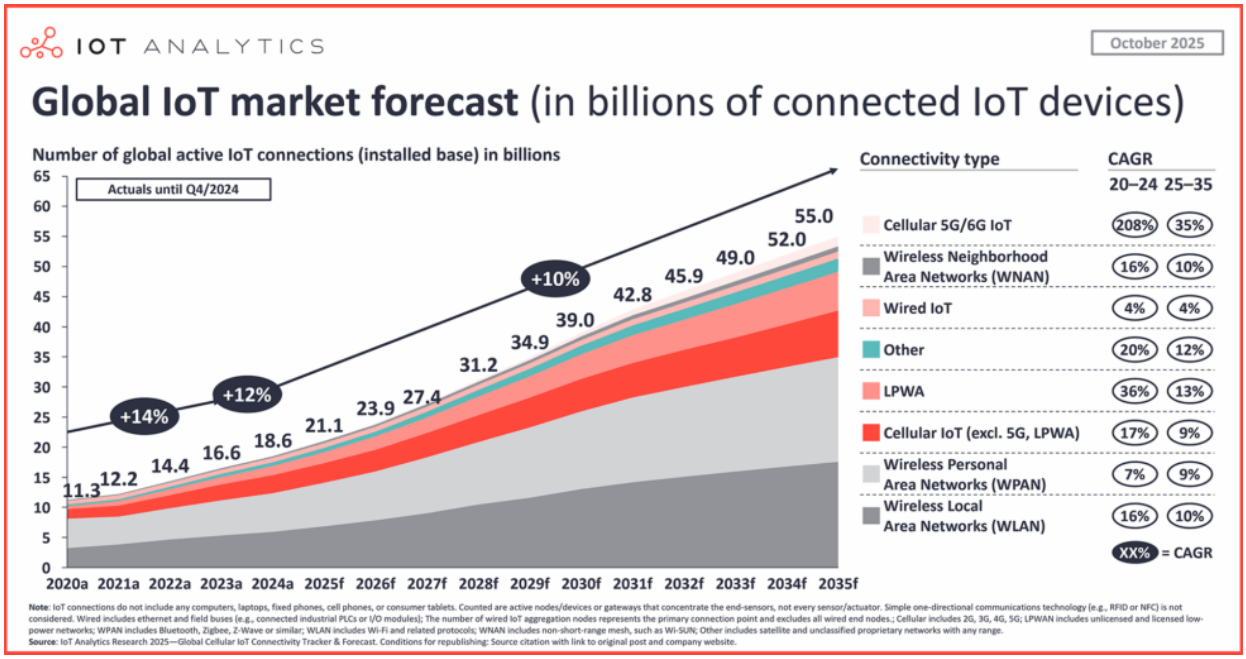


Figure 1. Global IoT market forecast, adapted from IoT Analytics [14].

1.B - Risks Posed by Insecure IoT Devices

Attackers actively target SOHO routers and connected devices because they often sit at the edge of the network and may be poorly monitored. Microsoft Threat Intelligence reported in April 2026 that Forest Blizzard, a Russian-linked threat actor, compromised SOHO routers and changed DNS settings to route victim traffic through attacker-controlled DNS infrastructure [9]. This type of attack matters because DNS manipulation can support surveillance, credential targeting, and adversary-in-the-middle activity.

Other public advisories show the same pattern. The FBI warned that end-of-life routers can be exploited by cybercriminal proxy services when vendors no longer provide security updates [4]. Joint government advisories have also described botnet operations that use compromised routers and IoT devices to support malicious activity [5]. These examples show that SOHO network infrastructure is not too small to matter; it is often attractive because it is widely deployed, inconsistently patched, and difficult for normal users to monitor.

IoT risk is not limited to the possibility that a single device becomes compromised. Many devices automatically expose services for setup, discovery, pairing, media streaming, file sharing, and app integration. Home users may assume that creating a separate SSID or VLAN solves the problem, but misconfiguration, shared services, unblocked routes, multicast behavior, and device-specific features may still allow visibility or reachability across boundaries. Modern SOHO networks therefore behave like small enterprise networks, with different device classes, trust levels, and security requirements.

2 - Motivation

2.A - Why Home Users Lack Visibility

The Microsoft SOHO router example helped frame the motivation for this project because it showed that home and small-office infrastructure can expose useful information to attackers when it is compromised or poorly monitored [9]. My own motivation began several years ago when I started becoming more serious about networking and building a home lab. I added a firewall, separated SSIDs, and moved IoT devices onto their own network segment so they would not share the same level of access as personal devices.

Before making those changes, I ran packet captures to see what was actually happening on the network. I was surprised by how much information was visible: hostnames, IP addresses, service discovery traffic, requests, broadcasts, and multicast messages. Some of this behavior is normal because home devices need discovery mechanisms for streaming, printing, file sharing, smart-home control, and other convenience features. However, the captures showed that the network was much noisier and more revealing than I expected. Smart-home research has also shown that local communication can reveal information about devices and user environments [6].

When I tried to reduce that exposure, the router became a bottleneck. Many consumer ISP routers, including the Xfinity-provided router in my environment, limit advanced network configuration. Since the router was tied to the internet service plan, replacing it or changing the setup was not as simple as buying a new device. This is a common SOHO problem: users may want stronger segmentation, better firewall control, or more visibility, but the equipment available to them is often designed for convenience instead of detailed security management.

This matters because routers are critical network infrastructure. They sit at the edge of homes, small businesses, schools, and remote-work environments. If a router or home network is misconfigured or compromised, it can enable traffic observation, DNS manipulation, botnet participation, unauthorized access, and attacks against connected devices. That experience motivated me to look for ways to measure what was visible and reachable on my own network instead of relying only on a router dashboard.

2.B - Need for a Free, Open-Source, and Adjustable Tool

The main gap I identified is that home users can often make network changes, but they may not know whether those changes actually reduced exposure. A router administration page may show that devices are connected to different SSIDs or VLANs, but it usually does not show whether mDNS, SSDP, or

selected TCP services are still visible across boundaries. A simple local measurement tool is useful because it tests the network from the user's own environment.

HOME-SHIELD is designed to fill that gap. Instead of assuming that segmentation works, the user can run a baseline scan, make a hardening change, run another scan, and compare the results. This makes home network hardening more evidence-based. The user can see whether responders disappeared, whether open ports closed, and whether any new exposure appeared after the change.

Another design goal is accessibility. Commercial enterprise monitoring platforms may be too expensive, too complex, or too heavy for a student, home lab, or small SOHO environment. HOME-SHIELD is meant to be free, lightweight, understandable, and adjustable. A beginner can use it to generate a report, while a more technical user can modify port lists, add protocol modules, tune the scoring model, or adapt the tool to a different network. Different users may prioritize different devices, such as smart TVs, speakers, cameras, NAS devices, or work-from-home systems, so the tool should provide a clear starting point without forcing every environment into the same model.

3 - Project Overview

3.A - What Is HOME-SHIELD?

HOME-SHIELD is a Python-based CLI tool that measures a network's exposure by observing which devices and services remain visible or reachable from a specific vantage point. It is a defensive tool, not an exploitation tool. I designed it to support repeatable network measurements rather than a one-time scan.

The tool gathers discovery and reachability data, compares two runs, generates an HTML report, and optionally analyzes packet captures for supporting evidence. Its central question is: did a change such as VLAN isolation, a firewall rule, service restriction, multicast adjustment, or routing change reduce observable exposure in the network? HOME-SHIELD can also function as a measurement framework for tracking exposure over time.

This project is not just a collection of ad hoc commands. It is designed as a structured measurement framework that produces repeatable evidence. That distinction is important because the goal is not only to see what exists on the network, but to compare what changed after hardening.

3.B - Measure -> Harden -> Re-measure Workflow

The central workflow is Measure -> Harden -> Re-measure. The first measurement establishes a baseline state. This baseline shows what is visible and reachable before any changes are made. The user can also run the baseline alone when they want a quick view of exposure from their current network position.

The hardening step introduces a controlled change. This may include tightening firewall rules, enabling client isolation, disabling unnecessary services, modifying VLAN policy, adjusting multicast forwarding behavior, or restricting selected ports. HOME-SHIELD intentionally does not perform the hardening itself; it measures the outcome of whatever hardening strategy the user applies.

The second measurement captures the post-change state. The diff stage translates two separate observations into meaningful deltas. Instead of manually comparing CSV files or screenshots, the tool identifies removed responders, newly added responders, closed ports, newly open ports, and score changes. This makes the result easier to interpret and defend.

3.C - Intended Users and Use Case

HOME-SHIELD is intended for technically capable home users, students, homelab operators, and small-network defenders who understand basic networking and want to validate segmentation or hardening changes. It is especially useful for anyone managing a SOHO environment where IoT devices, personal devices, and work systems share infrastructure.

The generated reports are designed to be understandable to readers with general technology knowledge. The tool preserves detailed outputs for technical review while also producing a readable summary for professors, reviewers, or other stakeholders.

4 - System Design, Architecture, and Implementation

4.A - Python CLI Implementation

Python was the strongest implementation choice for HOME-SHIELD because the project requires structured parsing, file generation, modular design, comparison logic, and maintainable code. Python also supports rapid development while remaining readable for review and troubleshooting.

A command-line interface was chosen because the workflow is procedural and repeatable. The user runs a command, stores artifacts, applies a change, runs another command, and compares the outputs. A CLI also supports future automation, such as scheduled measurements, integration with other lab tools, or long-term exposure history tracking.

4.B - Modular Architecture

HOME-SHIELD uses a modular architecture. The tool separates command routing, discovery, reachability testing, diffing, reporting, packet-capture parsing, and shared utilities. This design makes the tool easier to maintain and troubleshoot because each component has a clear responsibility.

The modular structure also improves testability. Discovery logic, reachability checks, diff calculations, report generation, and output handling can be validated separately. Finally, the architecture supports future upgrades. Additional protocols, new output formats, and revised scoring rules can be added without rewriting the entire tool.

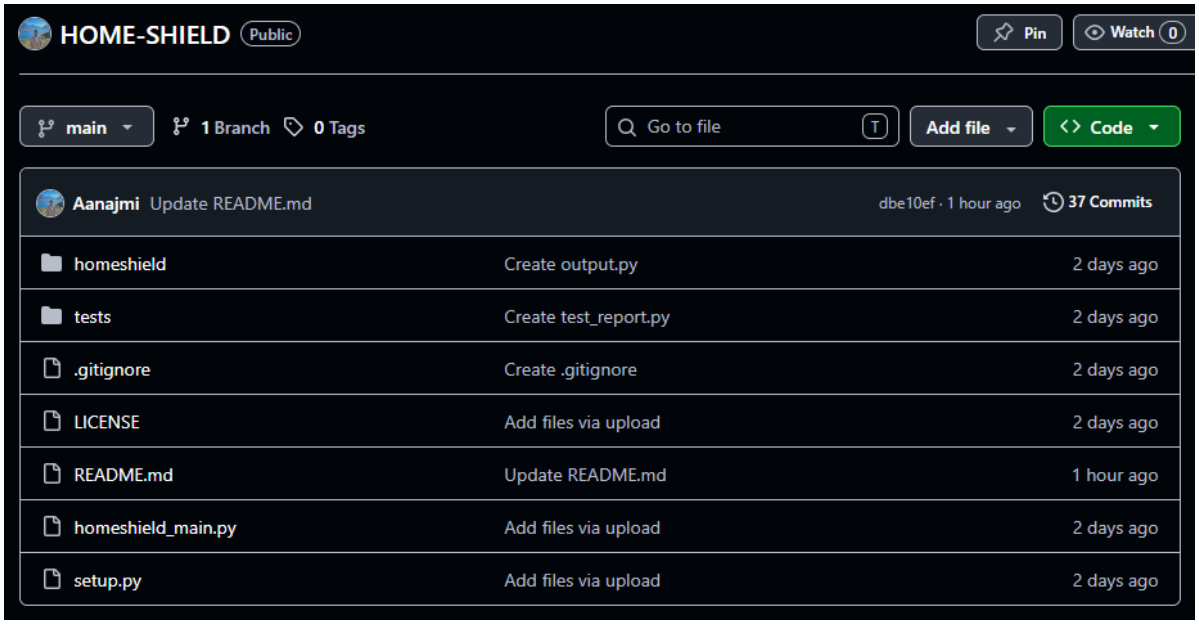


Figure 2. HOME-SHIELD GitHub repository structure.

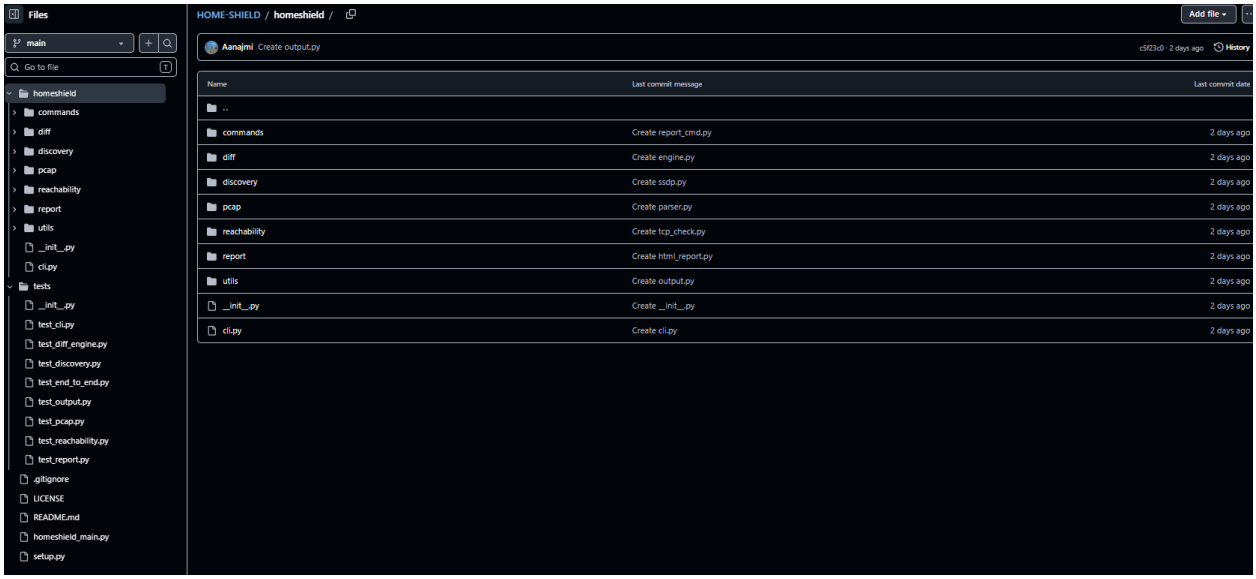


Figure 3. HOME-SHIELD package structure with separate modules for commands, discovery, diffing, PCAP parsing, reachability, reporting, and utilities.

4.C - Implemented Commands

HOME-SHIELD provides four main commands that match the project workflow. The commands are designed to keep measurement, comparison, reporting, and evidence review separate.

Command	Purpose
measure	Runs discovery checks and TCP reachability tests from the current vantage point.
diff	Compares two run.json files and identifies discovery, reachability, and score changes.
report	Generates a readable HTML report from a diff output.
pcap-parse	Optionally extracts mDNS/SSDP talkers from a packet capture using supporting packet-analysis tooling.
--help	Displays available commands and examples.

Example command patterns:

```
homeshield measure --label baseline_iot --vantage iot --iface en0 --rounds 5 --interval 10 --listen 4 --ports 80,443,445,554,1883
```

```
homeshield diff --before outputs/baseline_iot_run1/run.json --after outputs/rerun_iot_run3/run.json --out outputs/rerun_iot_run3_diff.json
```

```
homeshield report --diff outputs/rerun_iot_run3_diff.json --out reports/rerun_iot_run3_diff.html
```

```
homeshield pcap-parse --pcap evidence/pcaps/capture.pcap
```

```
abbasn@Abbass-Laptop ~ % homeshield --help
usage: homeshield [-h] [--version] {measure,diff,report,pcap-parse} ...

HomeShield — Home Network Exposure Measurement Tool

positional arguments:
  {measure,diff,report,pcap-parse}
                        Available commands
    measure             Run discovery + reachability and write outputs
    diff                Compare two run.json files (before/after)
    report              Generate HTML report from diff or run data
    pcap-parse          Extract mDNS/SSDP talkers from PCAP using tshark

options:
  -h, --help            show this help message and exit
  --version             show program's version number and exit

Examples:
  homeshield measure --label baseline_iot --vantage iot --iface en0 --rounds 5
  homeshield diff --before outputs/baseline/run.json --after outputs/hardened/run.json
  homeshield report --diff outputs/diff.json --out reports/report.html
  homeshield pcap-parse --pcap evidence/pcaps/capture.pcap
abbasn@Abbass-Laptop ~ %
```

Figure 4. HOME-SHIELD CLI help output showing supported commands.

4.D - Data Flow and Output Artifacts

HOME-SHIELD collects discovery responses and TCP reachability results, then writes structured output files. This allows users to inspect, compare, and preserve evidence. The output format is important because the tool is meant to support repeatable measurement rather than temporary terminal output.

Artifact	Purpose
run.json	Stores the full structured record of a measurement run.
discovery.csv	Provides a human-readable table of discovered devices and services.
reachability.csv	Provides a table of tested hosts, ports, and results.
diff.json	Records before-and-after changes and score movement.
HTML report	Converts technical evidence into a presentation-ready summary.

JSON supports automation and structured comparison, CSV supports manual inspection, and HTML supports evaluation and communication. This design supports both technical troubleshooting and academic reporting.

5 - Methodology

5.A - Discovery Approach Using mDNS and SSDP

HOME-SHIELD focuses on mDNS and SSDP because they are common discovery protocols in home and IoT environments. Multicast DNS lets devices on the same local network find names such as device.local without relying on a traditional DNS server. SSDP is used by UPnP-style devices to announce themselves and allow other devices to search for local services. These mechanisms are useful because consumer devices are expected to work with minimal setup.

From a security perspective, these protocols are valuable measurement targets because they reveal whether segmentation is limiting visibility. If a device on an IoT segment remains discoverable from a trusted LAN, or if a trusted device is visible from a guest network, then some discovery visibility exists across the boundary. Research on discovery protocols and smart-home communication shows that these local mechanisms can expose device and service information [6], [8].

In my packet captures, I observed mDNS PTR queries for services such as _sleep-proxy._udp.local, _meshcop._udp.local, _afpovertcp._tcp.local, and _smb._tcp.local. These names show that devices were actively looking for local services. That behavior is expected in many home networks, but it also supports the project's motivation: convenience features create observable network signals.

5.B - TCP Reachability Testing Approach

Discovery alone does not fully describe exposure. A device can be discoverable without exposing a reachable service, and a service can be reachable even if it was not discovered through mDNS or SSDP. For that reason, HOME-SHIELD also performs TCP reachability tests. These tests are not vulnerability scans; they simply determine whether a selected service is reachable enough to accept a connection attempt.

Result	Meaning
OPEN	The TCP connection succeeded, meaning the service is reachable from the scanner.
CLOSED	The target responded, but no service accepted the connection on that port.
TIMEOUT	The connection attempt did not receive a useful response, which may indicate filtering, packet loss, or an unavailable host.

These categories are simple, but they are useful for segmentation validation. A change from OPEN to CLOSED or TIMEOUT after hardening suggests reduced reachable service exposure from the tested vantage point.

5.C - Exposure Score

The exposure score helps summarize results quickly without replacing the raw evidence. HOME-SHIELD starts with a base score of 100, where a higher score means less observed exposure. Points are then deducted for discovery responders and reachable services. The scoring model is intentionally transparent so users and reviewers can understand why the score changed.

Exposure Item	Score Impact
Base score	100
Each unique mDNS responder	-2

Each unique SSDP responder	-2
Each open TCP port	-5
Each newly open TCP port after hardening	-8 extra

The score is useful for quick interpretation, but it does not replace device-level and port-level details. A score improvement should always be explained by the underlying changes, such as fewer responders, closed services, or the absence of new exposure.

5.D - Logging, Error Handling, and Output Formats

Network measurement is often inconsistent because device state, packet loss, multicast behavior, and firewall behavior can vary. Devices may sleep, services may respond intermittently, multicast packets may be missed, and routes may change. Because of this, HOME-SHIELD preserves structured output even when individual observations are not perfectly clean.

The tool's output structure supports troubleshooting and review. Technical data can be stored, compared later, and converted into a report. This makes the project stronger because the results are based on artifacts rather than only live terminal output.

5.E - Optional PCAP Support

The optional PCAP parsing feature adds evidence support for users who want to confirm what discovery traffic appeared on the wire. Packet captures are useful when investigating discovery behavior or explaining why a device appeared in one scan but not another. This feature is optional so that advanced evidence review is available without making the core workflow harder to use.

6 - Test Environment

6.A - Home Lab and Network Setup

The testing environment was my home lab. I designed the lab to represent a realistic SOHO network with multiple SSIDs, VLAN-style segmentation, a FortiWiFi 60F, an Xfinity ISP router, and a mix of IoT and non-IoT devices. The environment included consumer devices, smart-home behavior, and local discovery protocols.

The scanner ran on macOS using interface en0. This scanning position matters because HOME-SHIELD results are vantage-dependent. A scan from another network segment could produce different results. Therefore, the findings should be interpreted as what was visible and reachable from the system running the tool.

Lab Element	Description
Scanning host	macOS laptop
Scanning interface	en0
Scanner IP observed in logs	192.168.1.124
Network equipment	FortiWiFi 60F and Xfinity ISP router
Network design	Multiple SSIDs with IoT and non-IoT devices
Measurement style	Baseline run compared against later re-measurement

6.B - Devices, Segments, Ports, and Protocols

The lab tests focused on common discovery and service exposure indicators. HOME-SHIELD observed local discovery behavior and tested selected TCP ports that are realistic in home and IoT environments.

Port	Protocol/Use	Security Relevance
80	HTTP web interface	Common management or embedded web interface.
443	HTTPS web interface/API	Common encrypted web dashboard or service endpoint.
445	SMB file sharing	Can expose file-sharing services or Windows-style network access.
554	RTSP streaming	Common on cameras and media devices.
1883	MQTT IoT messaging	Common in IoT and home-automation messaging.

The goal was not to scan every possible port. The goal was to test a focused set of services that are meaningful in a home IoT exposure context.

6.C - Baseline and Hardening Scenarios

The final HOME-SHIELD evaluation compared two runs: baseline_iot_run1 and rerun_iot_run3. The baseline run represented the original observed exposure state. The re-measurement run represented the later state after additional testing and hardening work. The report showed changes across both discovery and reachability, with a clear score improvement tracked in the generated HTML report.

The generated report timestamp was 2026-04-27T15:33:45.060541+00:00. The run logs also show that the final measurement used five discovery rounds and tested six discovered IPs across five ports, producing 30 TCP reachability checks.

7 - Evaluation and Results

7.A - Evaluation Criteria

The evaluation focused on whether HOME-SHIELD could perform the expected workflow successfully and produce understandable evidence. The evaluation criteria were:

- Capture a baseline exposure state.
- Capture a hardened or later re-measurement state.
- Compare discovery and reachability changes.
- Produce a clear exposure score.
- Generate a readable report explaining whether exposure improved or worsened.

This evaluation matters because HOME-SHIELD is not only a scanner. Its main value is comparison: showing how exposure changes between two points in time.

7.B - Functional Verification

The main commands worked together as intended. The measure command captured network observations, the second measurement captured the post-change state, the diff command compared

both runs, and the report command created a readable verdict. The final report included a baseline score, re-measurement score, score change, discovery changes, reachability changes, and final verdict.

The final test report stated: "IMPROVED: Network exposure improved by 3 points. Removed 3 mDNS + 1 SSDP responders, 2 OPEN ports closed." This verified the core project workflow from measurement to comparison to reporting.

The HTML report was also tested for readability. In ten informal user trials, all ten users were able to understand the report's main outcome because it used clear labels, colors, and separated discovery from reachability results. This supported the usability goal of making technical results easier to interpret.

7.C - Before-and-After Results

The final scan showed a measurable improvement. The baseline exposure score was 55, and the re-measurement score was 58, producing a +3 improvement. Discovery visibility decreased because mDNS responders dropped from 9 to 6. Open TCP ports decreased from 5 to 4.

Metric	Baseline	Re-measurement	Change
Exposure score	55	58	+3
mDNS responders	9	6	-3
SSDP responders	1	1	0 net
Open TCP ports	5	4	-1

Although the SSDP count stayed the same, the actual SSDP device changed. One SSDP responder was removed and another was added. This is important because the same total count can hide device-level changes. HOME-SHIELD made that visible in the detailed report.

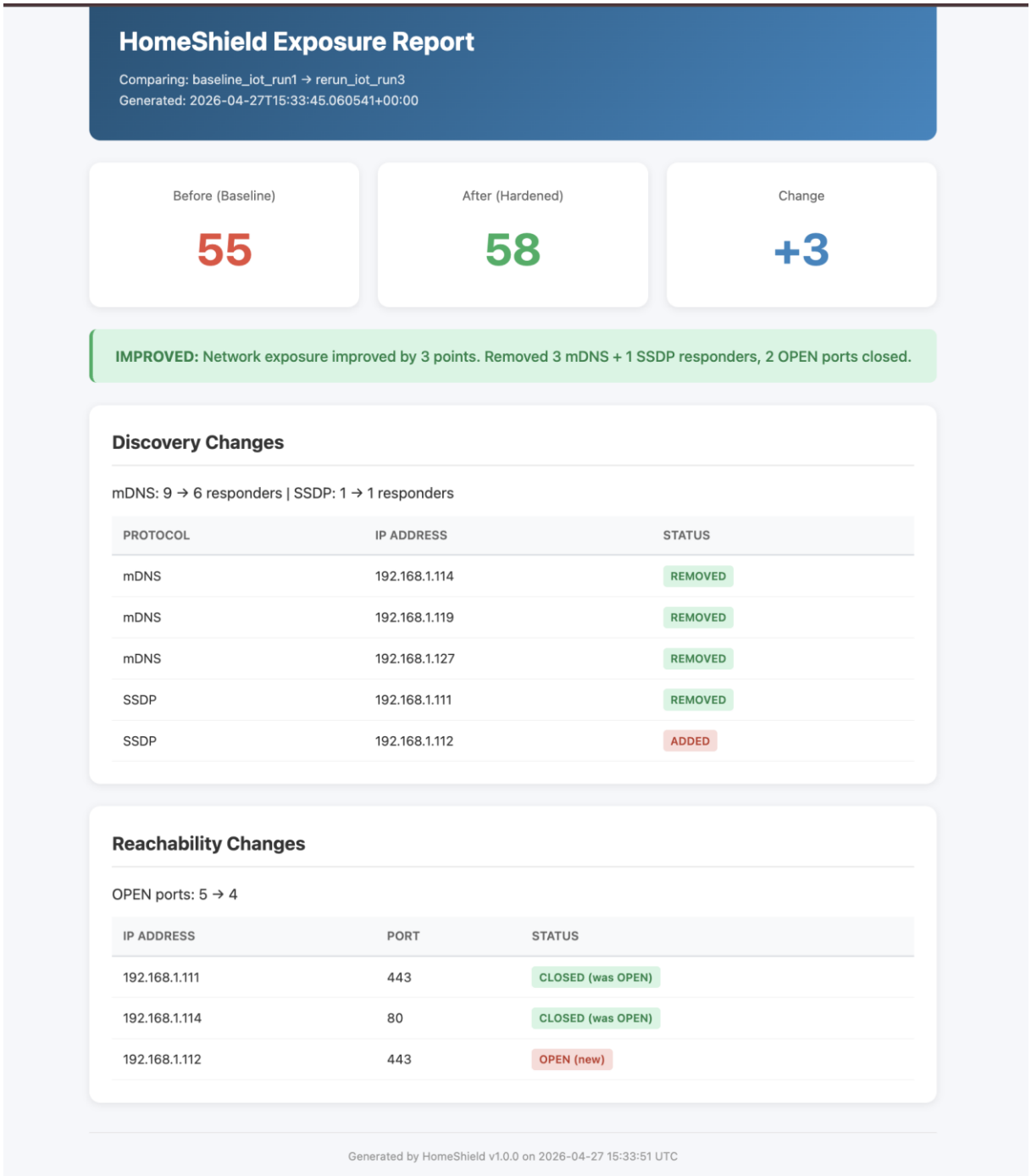
Protocol	IP Address	Change
mDNS	192.168.1.114	Removed
mDNS	192.168.1.119	Removed
mDNS	192.168.1.127	Removed
SSDP	192.168.1.111	Removed
SSDP	192.168.1.112	Added

The reachability results also changed. Two previously open services were closed, while one new HTTPS service appeared. The closed ports are important because they show reduced reachable service exposure. Port 443 is commonly used for HTTPS services, including dashboards and APIs. Port 80 is commonly used for HTTP web interfaces. In a home IoT environment, exposed web services can increase attack surface because they may provide management pages, application interfaces, or embedded web servers.

IP Address	Port	Baseline	Re-measurement	Security Meaning
192.168.1.111	443	OPEN	CLOSED	Reduced HTTPS exposure.
192.168.1.114	80	OPEN	CLOSED	Reduced HTTP exposure.
192.168.1.112	443	Not open	OPEN	New exposure to investigate.

The newly open port on 192.168.1.112:443 limited the score improvement. This is a useful finding because it shows that HOME-SHIELD does not simply report good news after hardening. It also identifies

regressions or new exposure that may need follow-up investigation. The network improved overall, but it was not a perfect hardening result.



HomeShield Exposure Report

Comparing: baseline_iot_run1 → rerun_iot_run3
Generated: 2026-04-27T15:33:45.060541+00:00

Before (Baseline)	After (Hardened)	Change
55	58	+3

IMPROVED: Network exposure improved by 3 points. Removed 3 mDNS + 1 SSDP responders, 2 OPEN ports closed.

Discovery Changes

mDNS: 9 → 6 responders | SSDP: 1 → 1 responders

PROTOCOL	IP ADDRESS	STATUS
mDNS	192.168.1.114	REMOVED
mDNS	192.168.1.119	REMOVED
mDNS	192.168.1.127	REMOVED
SSDP	192.168.1.111	REMOVED
SSDP	192.168.1.112	ADDED

Reachability Changes

OPEN ports: 5 → 4

IP ADDRESS	PORT	STATUS
192.168.1.111	443	CLOSED (was OPEN)
192.168.1.114	80	CLOSED (was OPEN)
192.168.1.112	443	OPEN (new)

Generated by HomeShield v1.0.0 on 2026-04-27 15:33:51 UTC

Figure 5. HOME-SHIELD final exposure report comparing baseline_iot_run1 to rerun_iot_run3.

8 - Discussion

8.A - What the Results Show

The final test shows that HOME-SHIELD successfully measured a meaningful before-and-after exposure change. The exposure score increased from 55 to 58, and the tool classified the later state as improved.

More importantly, the improvement was explainable. The report showed that three mDNS responders were removed, one SSDP responder was removed, and two open ports were closed.

This supports the main argument of the project: home network hardening should be measured, not assumed. A user can configure a VLAN, SSID, firewall rule, or other network change, but the important question is whether that change affects observable exposure. HOME-SHIELD helps answer that question by comparing baseline and re-measurement results.

The result also shows why detailed reporting matters. If the report only showed a score, the user would know that the network improved by three points but would not know why. By showing removed responders and changed ports, HOME-SHIELD makes the result easier to understand and defend.

8.B - Practical Value of the Tool

In my tests, the tool did not just say that exposure improved; it showed exactly which discovery responders disappeared and which ports changed state. This is useful for SOHO environments because router dashboards often do not show cross-segment exposure clearly. A router may show that a device is connected, but it may not show whether that device is discoverable from another segment or whether its web service is reachable.

HOME-SHIELD also helped identify unexpected findings. For example, a newly open port deserves follow-up. It could be intended behavior, or it may be an unwanted service that became reachable. Either way, the tool gives the user a concrete item to investigate.

9 - Limitations

9.A - Limits of Multicast-Based Discovery

HOME-SHIELD uses discovery protocols such as mDNS and SSDP because they are common in home and IoT networks, but not all devices respond consistently. Some devices may advertise services only when they are awake, recently used, connected to a phone or application, or actively communicating with another device.

A missing discovery response does not always prove that a device is gone or fully isolated. In my final scan, mDNS responders decreased from 9 to 6, which was positive because fewer devices were visible through multicast discovery. However, the test also showed discovery churn: one SSDP responder was removed while another was added. This indicates that discovery results can change even when the network is not perfectly static [6].

Multicast behavior also depends heavily on network equipment such as routers, firewalls, access points, VLAN settings, multicast forwarding, and mDNS reflection. A device can be isolated at the firewall level but still appear through discovery if multicast traffic is repeated or proxied. For these reasons, HOME-SHIELD treats discovery as an exposure indicator, not absolute proof of security or insecurity.

9.B - Limits of Port-Based Reachability Testing

Port-based reachability testing shows whether selected TCP ports are open, closed, or timed out from the scanner's vantage point. This is useful because it shows whether a service can be reached across the tested path. However, it does not provide a full vulnerability assessment.

For example, in my final scan, HOME-SHIELD showed that 192.168.1.111:443 and 192.168.1.114:80 changed from open to closed. This was a positive result because those services were no longer reachable from the scanner. However, there is no evidence that the devices are completely secure. The result only proves that those specific services were no longer reachable from the tested vantage point.

An open port also does not automatically mean that a device is vulnerable. A service may be intentionally allowed, properly authenticated, and required for normal use. At the same time, an open port still increases exposure because it gives another host something to connect to, fingerprint, or attempt to authenticate against.

9.C - Generalizability and Tooling Constraints

The results from this project should not be treated as universal results for every home network. Home networks vary widely depending on equipment, configuration, firmware versions, device types, and IoT ecosystems. A result observed in my lab may look different in another home with different equipment.

This matters because HOME-SHIELD is vantage-dependent. The report describes what was visible and reachable from the system running the tool. A scan from another segment could produce different results. HOME-SHIELD demonstrates feasibility and usefulness in a realistic home-lab environment, but it does not claim to produce universal baselines. Its strongest claim is that it provides defensible local findings for the tested network and vantage point.

10 - Future Work

10.A - Additional Protocols

One area for future work is adding support for more discovery and management protocols. The current version focuses on mDNS, SSDP, and selected TCP ports, which are useful for many home and IoT environments, but they do not cover every protocol that may reveal devices or services. Future versions could add LLMNR, NetBIOS name service, ARP-based local discovery, printer discovery, and other protocol-specific modules [8].

This would allow HOME-SHIELD to detect a wider range of devices and services. It would also make the tool more useful across different home environments because not every device relies on the same discovery method.

10.B - Better Visualization and Reporting

Another improvement is better visualization and reporting. The current HTML report already helps make results easier to understand by showing before-and-after scores, discovery changes, reachability changes, and an overall verdict. However, the reporting system could be expanded further.

Future reports could include charts showing exposure score history across multiple runs. They could also show per-device timelines, repeated open ports, newly discovered services, and recurring exposure patterns. This would be useful because home networks change over time as devices are added, removed, updated, or moved between segments.

Better visualization would make HOME-SHIELD easier to use for non-technical readers. A professor, homeowner, or reviewer may not want to inspect raw JSON or CSV files. A clearer report with graphs and trend summaries would make findings easier to explain.

10.C - Expanded Scoring and Classification

The scoring model is intentionally simple, but future versions could add more context. Not all open ports or discovery responses carry the same level of concern. A newly open web management interface may deserve more weight than an expected media service. A future version could classify findings by device type, service type, or expected policy.

Another improvement would be allowing the user to mark certain exposure as intended. Some services may be required for normal usability, such as casting or smart-home control. A policy-aware scoring model could distinguish intended connectivity from unexpected exposure.

10.D - Broader Deployment and Testing

This project was tested in my own home lab, which was useful for proving the concept and validating the workflow. Broader testing would make the project stronger. Windows operating system support is a key goal for a future version because many users run network tools from Windows hosts.

HOME-SHIELD should also be tested across more routers, wireless systems, IoT devices, firewall configurations, VLAN designs, and home network layouts. This would help identify false positives, false negatives, and cases where discovery or reachability results are affected by vendor-specific behavior.

Broader testing would also improve confidence in the scoring system. If the tool is tested across many environments, it would be easier to determine whether the score weights are reasonable or need adjustment.

11 - Conclusion

11.A - Summary of the Problem

Home networks are becoming more complex because they now include many IoT devices. These devices are convenient, but they often rely on local discovery, automatic pairing, streaming protocols, file-sharing services, and application control channels. This creates visibility and reachability inside the local network that users may not understand or monitor.

Many users improve security by creating a separate SSID, guest network, or VLAN for IoT devices. This is a good practice, but configuration alone does not prove that the network is properly isolated. A router may show that segmentation exists, but the user still needs to know whether devices and services remain visible or reachable across boundaries. The main problem is that consumer networks are configurable, but not always observable.

11.B - Summary of the Solution

HOME-SHIELD addresses this problem by providing a repeatable measurement workflow for SOHO networks. The tool follows a Measure -> Harden -> Re-measure process: the user captures a baseline, applies a hardening change, runs a second measurement, and compares the results.

The tool measures both discovery and reachability. Discovery testing identifies which devices or services are visible through protocols such as mDNS and SSDP. TCP reachability testing checks whether selected ports can be reached from the scanner's vantage point. The tool then generates structured output, calculates an exposure score, identifies changes, and produces a readable HTML report.

In my final test, HOME-SHIELD compared baseline_iot_run1 to rerun_iot_run3 after additional testing and hardening work. The exposure score improved from 55 to 58. The report showed that three mDNS responders were removed, one SSDP responder was removed, and two previously open ports were closed. The test also found one newly open port, which showed that the tool can report both improvements and new exposure.

11.C - Final Takeaways

Home network hardening should be measured, not assumed. Creating a VLAN, separate SSID, or firewall rule is only part of the security process. The next step is verifying whether the change reduced

observable exposure. HOME-SHIELD supports that goal by giving users a practical way to compare before-and-after network states.

HOME-SHIELD is not a vulnerability scanner. Instead, it focuses on determining what devices and services are visible or reachable from a specific network vantage point. The project also showed that real home network measurements can be messy. Discovery protocols may vary between runs, and new services may appear while others disappear. This makes measurement even more important.

Overall, HOME-SHIELD demonstrates that home network exposure can be measured in a repeatable and understandable way. The tool combines cybersecurity concepts, networking knowledge, Python implementation, and readable reporting into one practical project.

For more information on HOME-SHIELD and how to install it, please visit:
https://github.com/Aanajmi/HOME-SHIELD

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Appendix

Appendix A - CLI Run and Report Generation Evidence

The screenshot below shows HOME-SHIELD being run from the command line, including the final measurement, diff generation, and HTML report generation. It supports the evaluation by showing that the tool completed the end-to-end workflow.

```
abbasn@Abbass-Laptop homeshield_submission % python3 homeshield_main.py measure --label rerun_iot_run3 --vantage iot --iface en0 --rounds
10:31:35 | INFO     | HomeShield logging initialized | log file: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/homeshield
10:31:35 | INFO     | HomeShield v1.0.0 | command: measure
10:31:35 | INFO     | ============================================================
10:31:35 | INFO     | MEASURE START: label=rerun_iot_run3, vantage=iot
10:31:35 | INFO     | ============================================================
10:31:35 | WARNING  | ip command fallback failed for en0: [Errno 2] No such file or directory: 'ip'
10:31:35 | INFO     | Interface en0 resolved to IP 192.168.1.124
10:31:35 | INFO     | Phase 1: Running discovery probes...
10:31:35 | INFO     | Starting mDNS discovery: interface_ip=192.168.1.124, listen=4.0s, rounds=5, interval=10.0s
10:31:35 | INFO     | mDNS round 1/5
10:31:39 | INFO     | mDNS round 1: 6 responders found
10:31:49 | INFO     | mDNS round 2/5
10:31:53 | INFO     | mDNS round 2: 5 responders found
10:32:03 | INFO     | mDNS round 3/5
10:32:07 | INFO     | mDNS round 3: 5 responders found
10:32:17 | INFO     | mDNS round 4/5
10:32:21 | INFO     | mDNS round 4: 6 responders found
10:32:31 | INFO     | mDNS round 5/5
10:32:36 | INFO     | mDNS round 5: 6 responders found
10:32:36 | INFO     | mDNS discovery complete: 6 unique responders
10:32:36 | INFO     | Starting SSDP discovery: interface_ip=192.168.1.124, listen=4.0s, rounds=5, interval=10.0s
10:32:36 | INFO     | SSDP round 1/5
10:32:40 | INFO     | SSDP round 1: 0 responders found
10:32:50 | INFO     | SSDP round 2/5
10:32:54 | INFO     | SSDP round 2: 1 responders found
10:33:04 | INFO     | SSDP round 3/5
10:33:08 | INFO     | SSDP round 3: 1 responders found
10:33:18 | INFO     | SSDP round 4/5
10:33:23 | INFO     | SSDP round 4: 1 responders found
10:33:33 | INFO     | SSDP round 5/5
10:33:37 | INFO     | SSDP round 5: 1 responders found
10:33:37 | INFO     | SSDP discovery complete: 1 unique responders
10:33:37 | INFO     | Total unique discovered IPs: 6
10:33:37 | INFO     | Phase 2: Running TCP reachability checks...
10:33:37 | INFO     | Starting reachability checks: 6 IPs x 5 ports = 30 checks (workers=10, timeout=1.0s)
10:33:40 | INFO     | Reachability checks complete: 30 total, 4 OPEN, 0 CLOSED, 26 TIMEOUT
10:33:40 | INFO     | Phase 3: Writing output artifacts...
10:33:40 | INFO     | JSON written: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/outputs/rerun_iot_run3/run.json (6880
10:33:40 | INFO     | CSV written: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/outputs/rerun_iot_run3/discovery.csv (
10:33:40 | INFO     | CSV written: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/outputs/rerun_iot_run3/reachability.cs
10:33:40 | INFO     | ============================================================
10:33:40 | INFO     | MEASURE COMPLETE: outputs in outputs/rerun_iot_run3
10:33:40 | INFO     | ============================================================
abbasn@Abbass-Laptop homeshield_submission % python3 homeshield_main.py diff --before outputs/baseline_iot_run1/run.json --after outputs/r
10:33:45 | INFO     | HomeShield logging initialized | log file: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/homeshield
10:33:45 | INFO     | HomeShield v1.0.0 | command: diff
10:33:45 | INFO     | ============================================================
10:33:45 | INFO     | DIFF START: before=outputs/baseline_iot_run1/run.json, after=outputs/rerun_iot_run3/run.json
10:33:45 | INFO     | ============================================================
10:33:45 | INFO     | JSON loaded: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/outputs/baseline_iot_run1/run.jsc
10:33:45 | INFO     | Loaded before run: baseline_iot_run1
10:33:45 | INFO     | JSON loaded: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/outputs/rerun_iot_run3/run.json
10:33:45 | INFO     | Loaded after run: rerun_iot_run3
10:33:45 | INFO     | Computing diff: before=baseline_iot_run1, after=rerun_iot_run3
10:33:45 | INFO     | Diff computed: before_score=55, after_score=58, improvement=+3
10:33:45 | INFO     | JSON written: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/outputs/rerun_iot_run3_diff.jsor (115
10:33:45 | INFO     | Diff output written: outputs/rerun_iot_run3_diff.json
10:33:45 | INFO     | ============================================================
10:33:45 | INFO     | DIFF COMPLETE: outputs/rerun_iot_run3_diff.json
10:33:45 | INFO     | ============================================================
abbasn@Abbass-Laptop homeshield_submission % python3 homeshield_main.py report --diff outputs/rerun_iot_run3_diff.json --out reports/rerun
10:33:51 | INFO     | HomeShield logging initialized | log file: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/homeshield
10:33:51 | INFO     | HomeShield v1.0.0 | command: report
10:33:51 | INFO     | ============================================================
10:33:51 | INFO     | REPORT START
10:33:51 | INFO     | ============================================================
10:33:51 | INFO     | JSON loaded: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/outputs/rerun_iot_run3_diff.json
10:33:51 | INFO     | Generating diff HTML report: reports/rerun_iot_run3_diff.html
10:33:51 | INFO     | HTML diff report written: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/reports/rerun_iot_run3_di
10:33:51 | INFO     | Diff report generated: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/reports/rerun_iot_run3_diff.
10:33:51 | INFO     | ============================================================
10:33:51 | INFO     | REPORT COMPLETE: /Users/abbasn/Downloads/s_jura-attachments/homeshield_submission/reports/rerun_iot_run3_diff.html
10:33:51 | INFO     | ============================================================
abbasn@Abbass-Laptop homeshield_submission % open reports/rerun_iot_run3_diff.html
abbasn@Abbass-Laptop homeshield_submission %
```

Figure A1. CLI evidence showing measure, diff, and report execution for the final run.

Appendix B - Installation and Project Repository

Project repository: https://github.com/Aanajmi/HOME-SHIELD